

EXECUTIVE BRIEFING

Cut your agentic-coding token bill

Coding agents are becoming a top-line AI cost. The lever is choosing the right **model and harness** from evidence -- not defaulting to the most expensive option.

WHAT YOU'LL GET

Three outcomes

A buying guide -- a cost/quality frontier that picks the model by evidence, not reputation.

A hidden lever -- the right agent cuts cost up to 3x on the same model, for the same result.

Managed spend -- route each task to the right tier and cut the bill with little quality give-up.

THE PROBLEM

Organizations are heading toward trillions of tokens a month -- and coding agents drive a large, growing share of it.

Two choices decide most of that bill -- **which agent (harness)** drives the work and **which model** it uses -- yet they are usually made on gut feel or on public leaderboards that may not reflect your real work.

WHY GENERIC NUMBERS MISLEAD

Real coding work doesn't look like a benchmark prompt

50-300

LLM turns per task -- long-horizon, not one-shot

150:1+

input-to-output ratio -- read-heavy, up to 660:1

3x

cost swing on the same model, just from the harness

Cost depends on **how** a model drives the task. You need numbers measured on **work that looks like yours**.

THE APPROACH

Measure **quality and **cost** on real coding tasks -- across agents and hosting options -- then let the **cost/quality frontier** pick the model.**

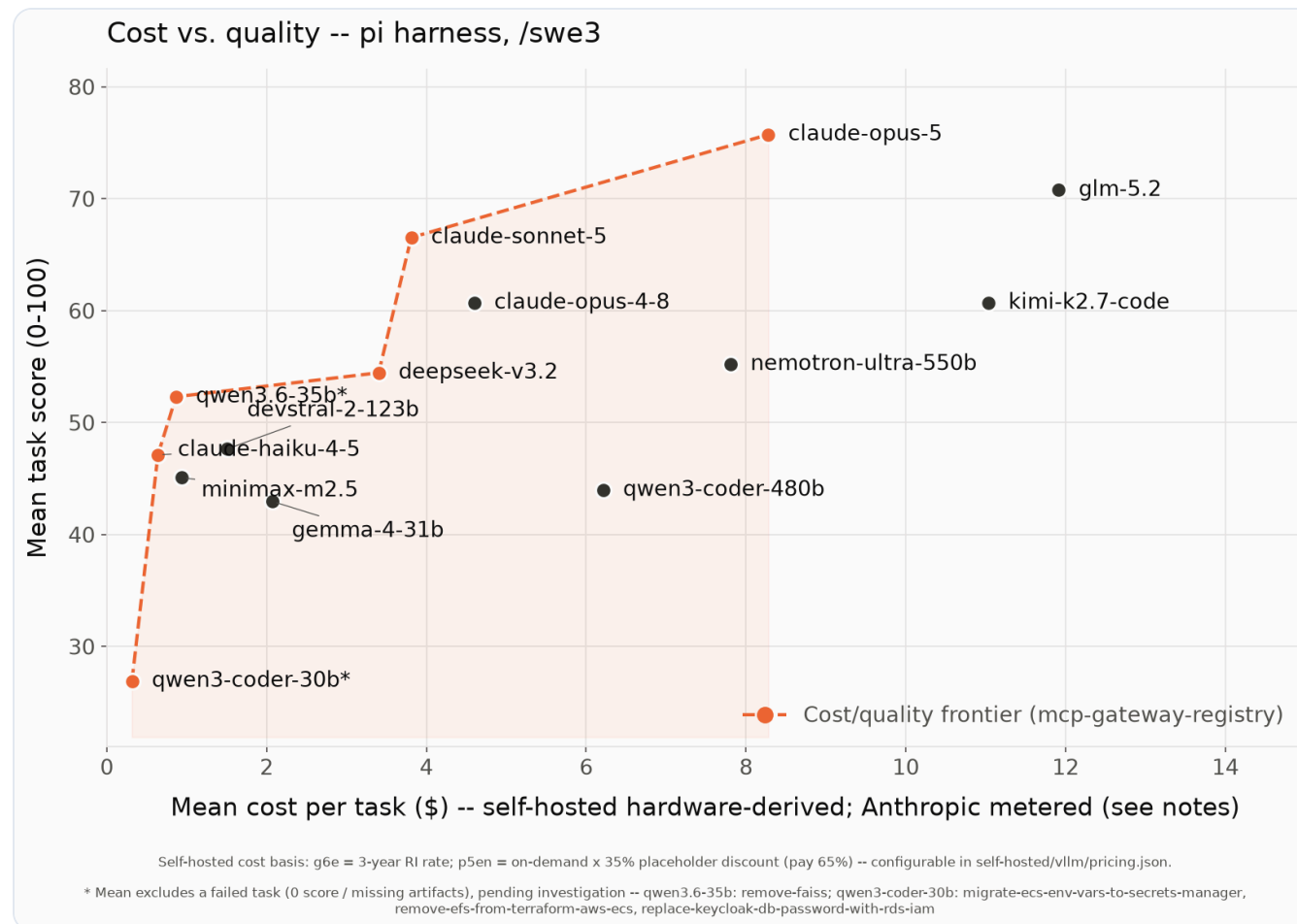
The frontier is the set of models where nothing else is both better and cheaper.
It turns model selection into a defensible, budget-aware decision.

HOW TO READ THE NEXT SLIDE

The **frontier** = the models where nothing else is both better *and* cheaper.

Everything off it is **dominated** -- some frontier model wins on both quality and cost, so you would never pick the other. To choose: pick the quality your task needs, then take the cheapest model on the frontier at that level. **Example:** a 71-point model at \$11.9/task is dominated by a 76-point model at \$8.3/task -- higher score, lower cost -- so it is off the frontier.

Your buying guide: the cost/quality frontier



14 models on real tasks. Higher is better; left is cheaper. The line is the frontier -- the only models worth considering. Everything below it is dominated (something is both better and cheaper).

WHAT TO BUY FOR WHAT

Match the model to the job

Business-critical work -- top-quality frontier model. Pay the most, get the most; use it where a wrong result is expensive.

Everyday engineering -- a workhorse model at a fraction of the cost delivers most of the quality for the bulk of tickets.

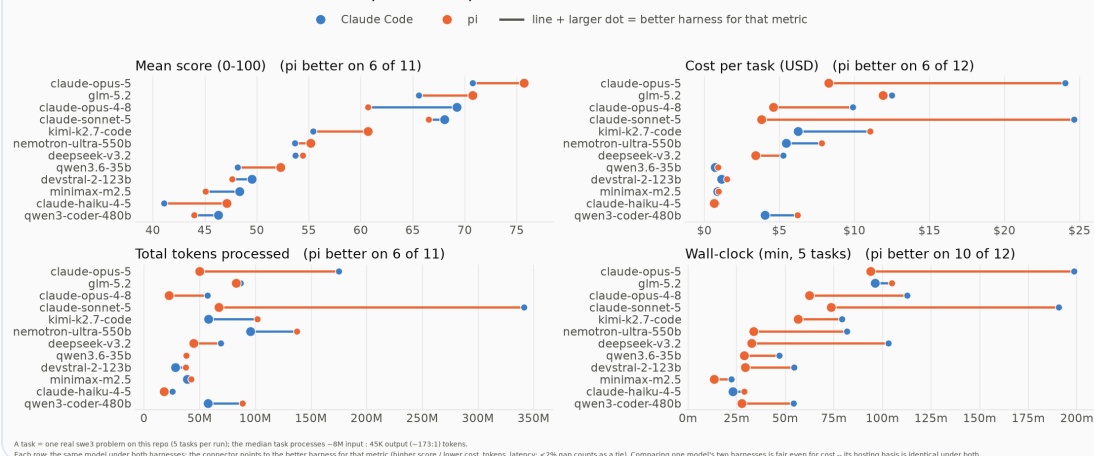
Routine / high-volume edits -- the most cost-effective model handles boilerplate and small fixes you review anyway.

The endgame: a harness that **routes each task to the right tier automatically** -- frontier quality at a fraction of the spend, with no one managing model choice.

THE OVERLOOKED LEVER

The agent matters as much as the model

Does the harness matter? Claude Code vs pi on swe3, per model (12 run under both)



Same model, different agent = up to 3× the cost and double the wall-clock, for little or no quality gain.

A leaner agent won faster turnaround on 10 of 12 models -- and tied or beat on cost.

Buy optionality: the freedom to pair each model with the cheaper agent is itself a cost lever.

BEDROCK OR SELF-HOSTED

**Works whether you buy tokens on
Amazon Bedrock or self-host open-
weight models on your own GPUs.**

Same tasks, same scoring -- so the trade-off is measured on one footing. At high volume, self-hosting a right-sized open-weight model on committed GPU capacity is a real lever against per-token pricing.

THE TAKEAWAY

Turn coding-agent spend into a managed decision

Stop defaulting to the most expensive model-and-agent for every task.

Choose from the frontier -- measured on your repos, your tasks, your hosting.

Route by tier -- and capture a large cost reduction with little quality give-up.

EXPLORE THE EVIDENCE

See the full benchmark

Open source. Reproduce the frontier on your own code and models.

github.com/aarora79/agentic-coding-harness-benchmarks



Scan for the repository